

VAJRALA GEETHIKA

Associate Software Engineer

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Summary

Detail-oriented Associate Software Engineer with a solid foundation in Python, SQL, and Agile methodologies. Collaborates effectively with cross-functional teams to design, test, and deploy software applications that solve complex problems. Experienced in all phases of the Software Development Life Cycle (SDLC), including structured coding, comprehensive debugging, and automated testing. Skilled in conducting system analysis, creating technical documentation, and delivering performance-optimized software solutions.

Technical Skills

Programming Languages:	Python, SQL, MySQL, PostgreSQL, C++
Development Frameworks:	Pandas, NumPy, Scikit-Learn, OpenCV
Tools & Platforms:	Git, GitHub, VS Code, Jupyter Notebook, Microsoft Excel
Software Engineering:	Software Development Life Cycle (SDLC), Agile, Scrum, Testing, Debugging

PROJECTS

Loan Default Risk Analysis Pipeline

Python, Pandas, SQL, ML

- Developed robust data processing and validation pipelines using Python and Pandas to analyze over 50,000 structured customer records.
- Implemented statistical testing methods using Scikit-Learn to improve software prediction accuracy metrics by exactly 15%.
- Collaborated with team members to establish design requirements and execute comprehensive code testing protocols.
- Created detailed user documentation and design specifications to ensure software alignment with quality standards.
- Debugged code anomalies and resolved execution bottlenecks to maintain overall system performance and consistency.
- Delivered weekly technical progress reports and software analysis updates to key project stakeholders.

Customer Purchase Behavior Analytics

SQL, Power BI, Excel

- Built optimized SQL queries to query, extract, and clean transaction records from relational databases hosting over 3,000 entries.
- Designed and tested interactive visualizations to track daily purchasing patterns and system KPIs in Scrum sprints.
- Automated manual data validation workflows in Python, which successfully reduced processing time by exactly 25%.
- Participated in regular code reviews and updated visual layouts based on collaborative team feedback.
- Documented database schemas, query instructions, and software setup guidelines for internal engineering use.
- Assisted in analyzing existing system processes to recommend software and database performance improvements.

AI-Powered Computer Vision Attendance System

Python, MySQL, OpenCV

- Developed a facial recognition pipeline in Python, testing and tuning image preprocessing to improve validation accuracy by exactly 25%.
- Implemented structured SQL schemas in MySQL to securely store, organize, and retrieve 1,000+ real-time logs.
- Tested database query performance and optimized execution times to sub-second levels to ensure consistency.
- Created automated Python scripts to generate clean, CSV-formatted reporting outputs for daily system checks.
- Collaborated with peer developers to integrate software modules and conduct regression testing on new updates.
- Maintained clear documentation of system architecture, installation steps, and code functionality parameters.

EDUCATION

SASTRA Deemed University | B. Tech in Electronics and Communication Engineering

Thanjavur, Tamil Nadu

CERTIFICATIONS

Microsoft Certified: Python for Data Science

Scaler Certified: SQL & Database Design with MySQL